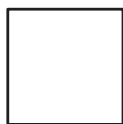
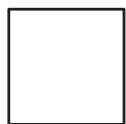
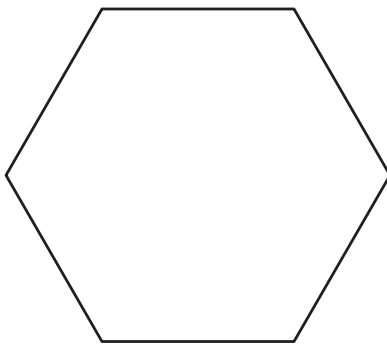
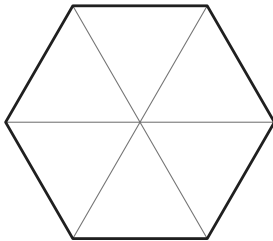
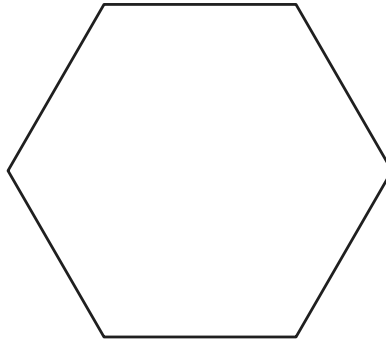
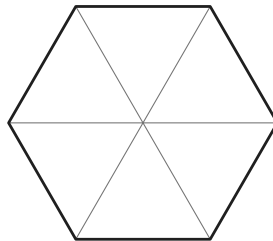
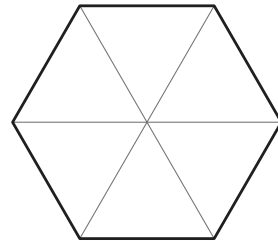


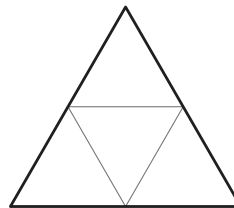
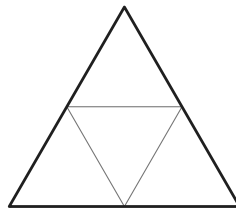
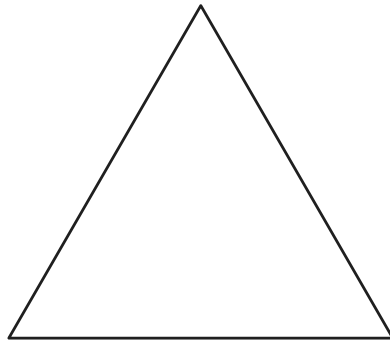
Problem 1: Cover the hexagon with green triangles only. Every part must be covered, with no gaps, no overlaps and nothing sticking out. Count the green triangles and tell the adult the number for the box with the triangle. Now cover the hexagon with blue rhombi only. Count them for the box with the rhombus. Now cover it with red trapezoids only. Count them for the box with the trapezoid.



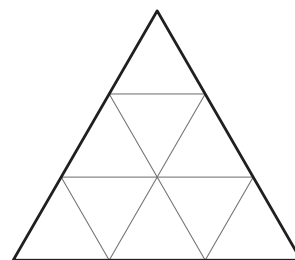
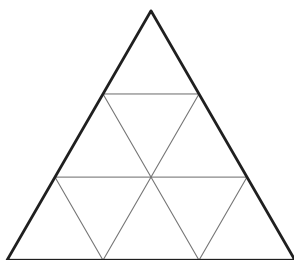
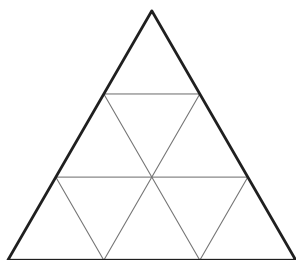
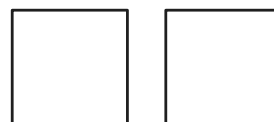
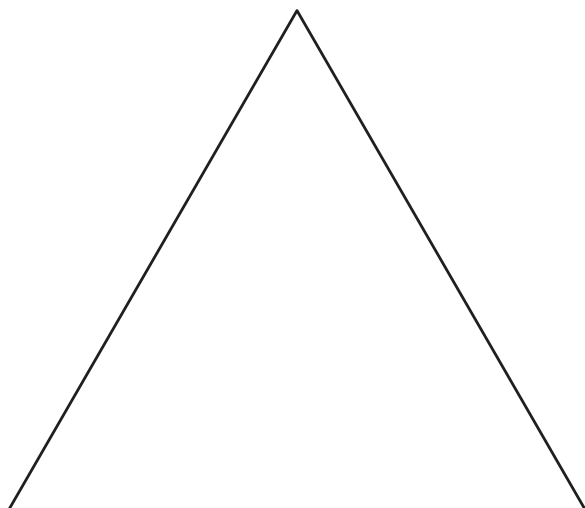
Problem 2: Cover the hexagon with one red trapezoid, one blue rhombus and one green triangle. Colour the small hexagon marked 3 to show where each piece went. Now cover the hexagon with exactly 4 pieces. You choose the colours. Colour the small hexagon marked 4. Now cover the hexagon with exactly 5 pieces. Colour the small hexagon marked 5.

**3****4****5**

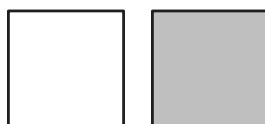
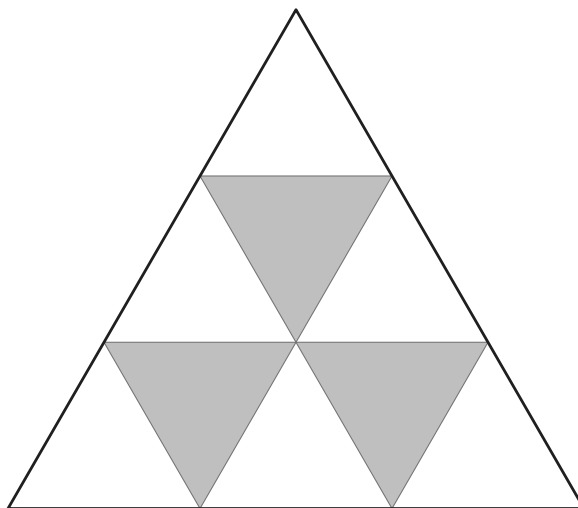
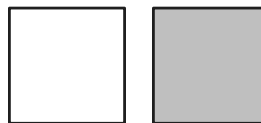
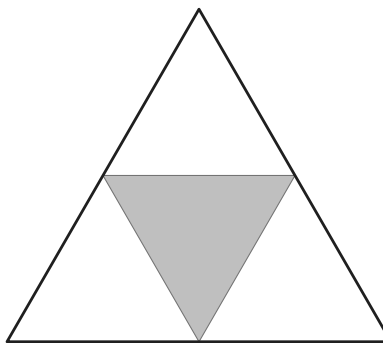
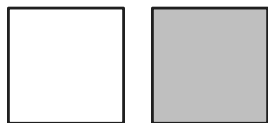
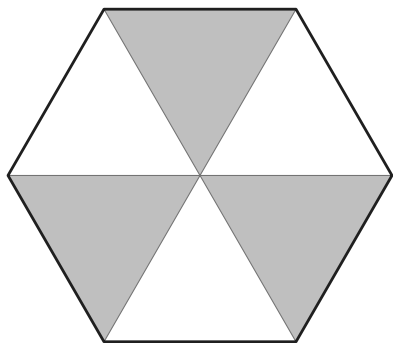
Problem 3: Cover the big triangle with green triangles only. Count them and tell the adult the number for the box. Now cover the big triangle with one red trapezoid and some green triangles. Colour the first small triangle to show where the red went. Now try to cover the big triangle with blue rhombi only. If you can, colour the second small triangle blue. If you cannot, cross it out and show the adult where a blue rhombus always gets stuck.



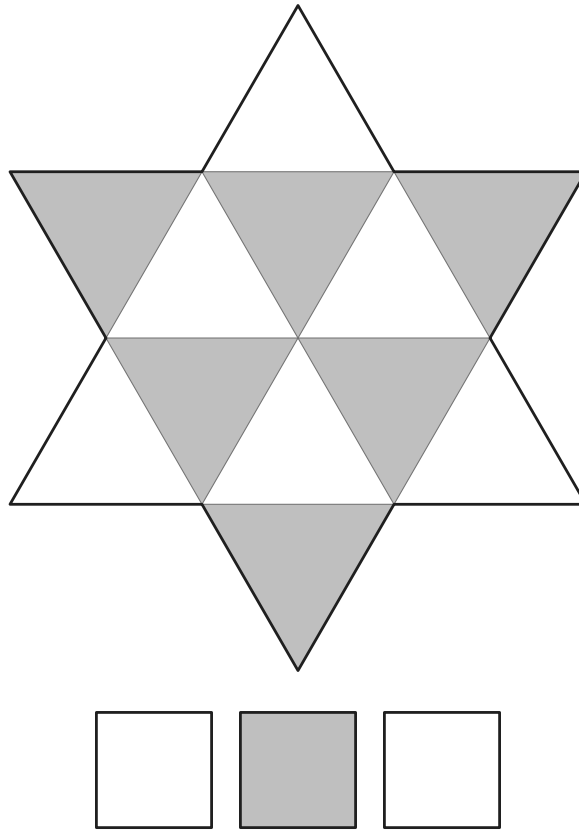
Problem 4: Cover the bigger triangle with green triangles only. Count them and tell the adult the number for the first box. Now cover it with red trapezoids only. Count them for the second box. Colour the first small triangle to show where the reds went. Now put the yellow hexagon inside the bigger triangle and cover the rest with green triangles. Colour the second small triangle to show where the yellow went. Now try to cover the bigger triangle with blue rhombi only. If you cannot, cross out the third small triangle and tell the adult how many triangles are left over.



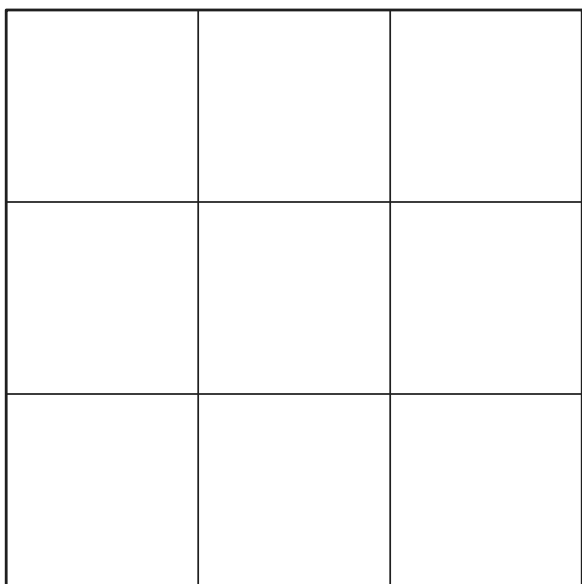
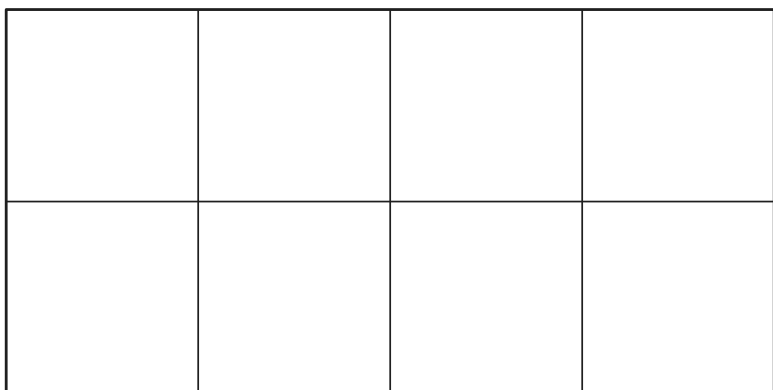
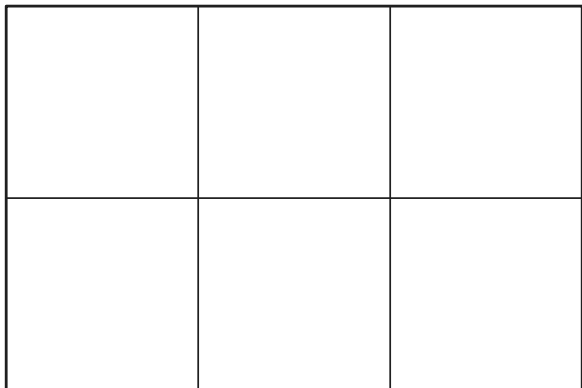
Problem 5: Here are the hexagon, the two triangles and a star, with some of the small triangles shaded grey. Put one blue rhombus anywhere on the shaded bigger triangle. Look at the two small triangles under it. What colour is each one? Move the blue rhombus to a new place and look again. Do this in six places. Now count the white triangles and the grey triangles in each shaded shape. Tell the adult the numbers for the white box and the grey box under each shape.



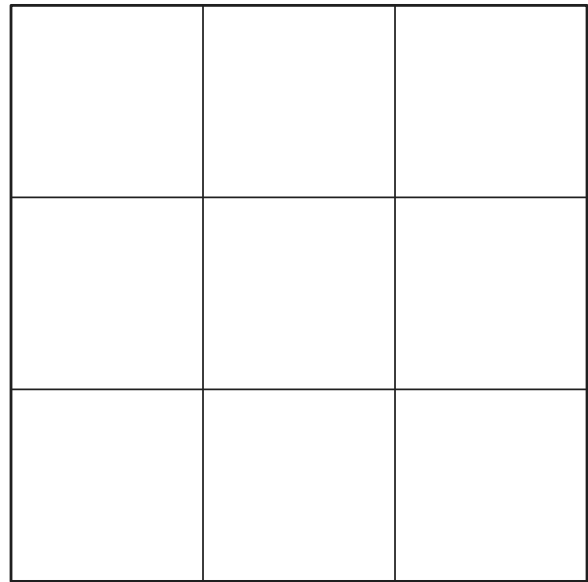
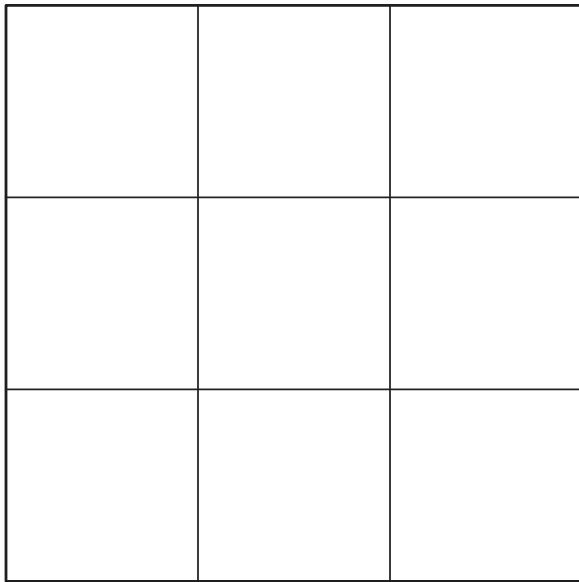
Problem 5 (continued): Then cover the shaded star with blue rhombi only and count them for its third box.



Problem 6: A domino covers two squares. Cover the 2-by-3 board with dominoes: every square covered, no overlaps, nothing sticking out. Count the dominoes and tell the adult the number for the box. Now cover the 2-by-4 board and count. Now try to cover the 3-by-3 board. Tell the adult how many squares you could not cover.

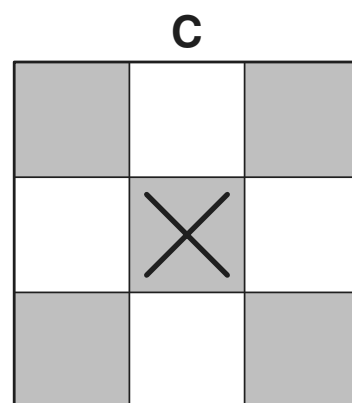
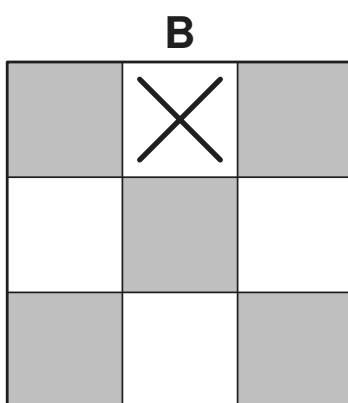
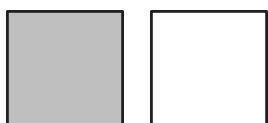
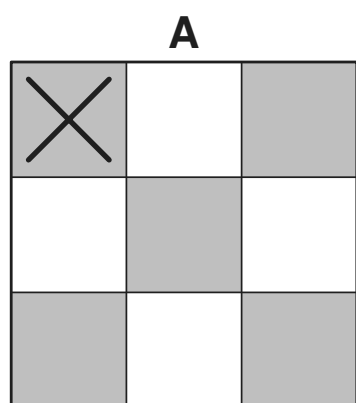
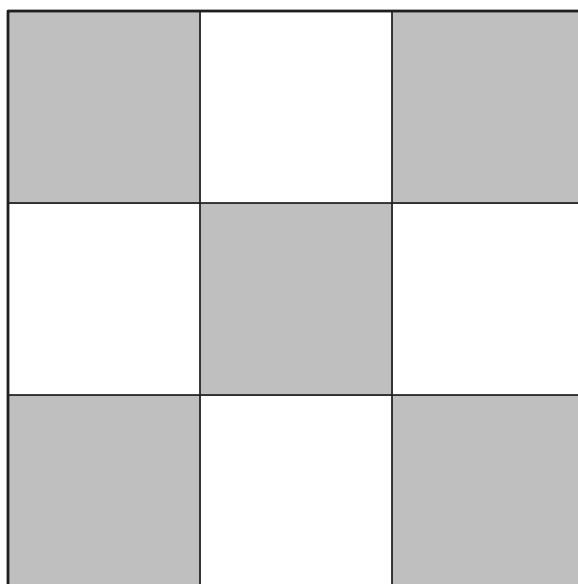


Problem 7: Put the X square on one square of the 3-by-3 board. That square is crossed out and may not be covered. Try to cover the other eight squares with four dominoes. If you can, draw a tick in the same square of the result grid. If you cannot, draw a cross there. Move the X square to a different square and try again. Do this for all nine squares. Look at the pattern of ticks and crosses.

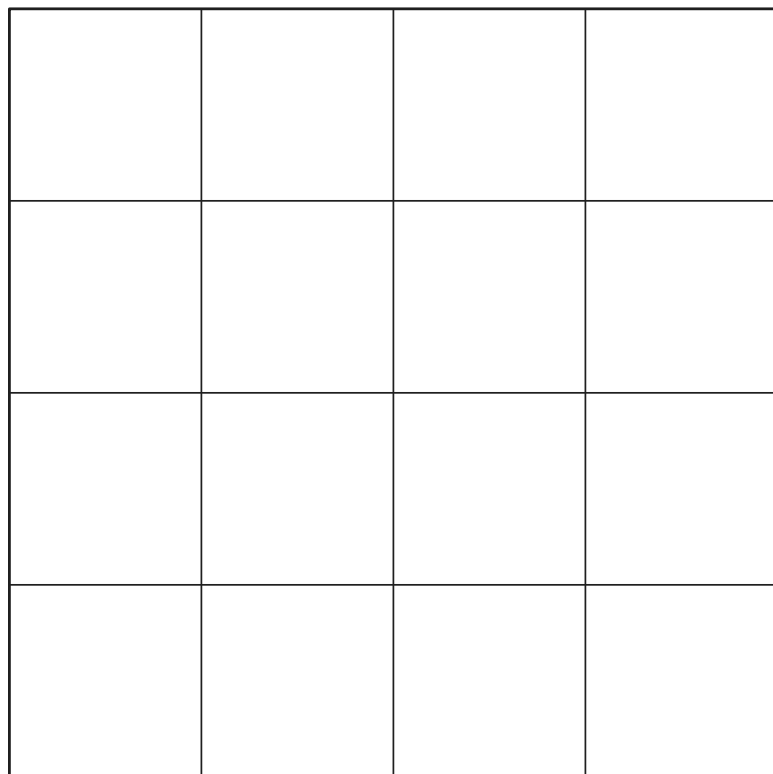
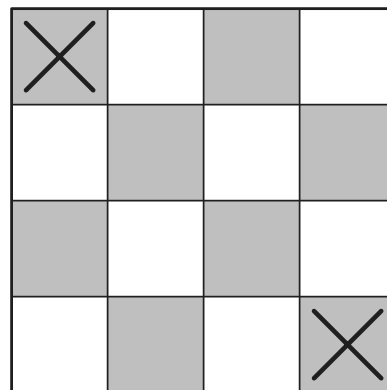


result grid

Problem 8: This 3-by-3 board is shaded like a checkerboard. Put one domino anywhere on it. Look at the two squares under it. What colour is each one? Move the domino to a new place and look again. Do this in five places. Now look at the three small boards. On each one, count the dark squares and the light squares that are not crossed out. Tell the adult the numbers for the dark box and the light box.



Problem 9: Put two X squares on the 4-by-4 board on the two corners shown in picture A. Try to cover the other 14 squares with seven dominoes. If you cannot, cross out picture A. Now move the X squares to the two corners shown in picture B and try again. If you cannot, cross out picture B. Count the dark squares and the light squares that are not crossed out in each picture and tell the adult the numbers for the boxes.

**A****B**